

Section 14.2: Limits and Continuity

Partial Derivatives

MTH 310
Calculus III

Limits in Multiple Variables

In Calc I, we asked: what happens to $f(x)$ as $x \rightarrow a$?

There were only **two directions** to approach from — left and right.

In Calc III, we ask: what happens to $f(x, y)$ as $(x, y) \rightarrow (a, b)$?

Now there are **infinitely many paths** to approach from — along lines, curves, spirals, any path you can imagine.

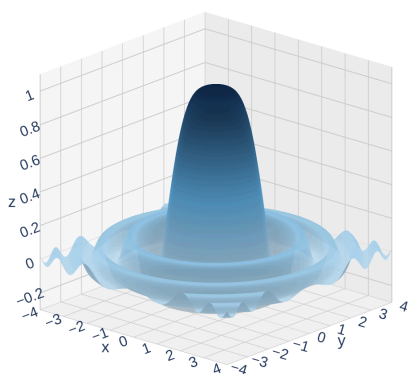
This is the key difference: in 1D you check two directions. In 2D you must worry about **every possible path** to the point.

Two Functions, Two Very Different Behaviors

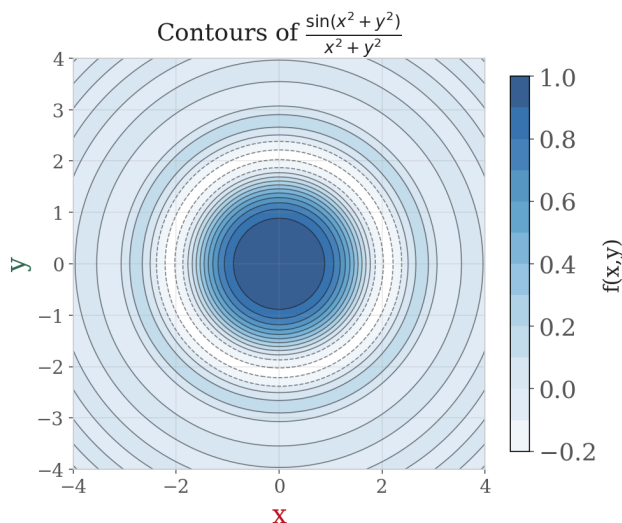
Let's compare what happens at the origin for two functions that both have the form $\frac{0}{0}$ there.

$$f(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$$

$$f(x, y) = \sin(x^2 + y^2) / (x^2 + y^2)$$



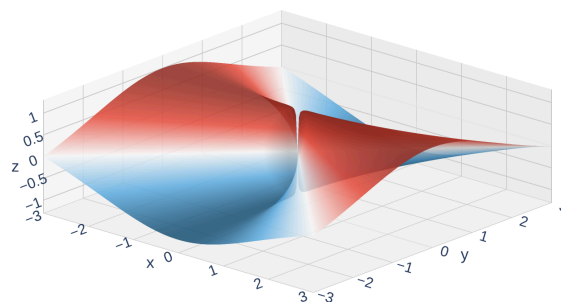
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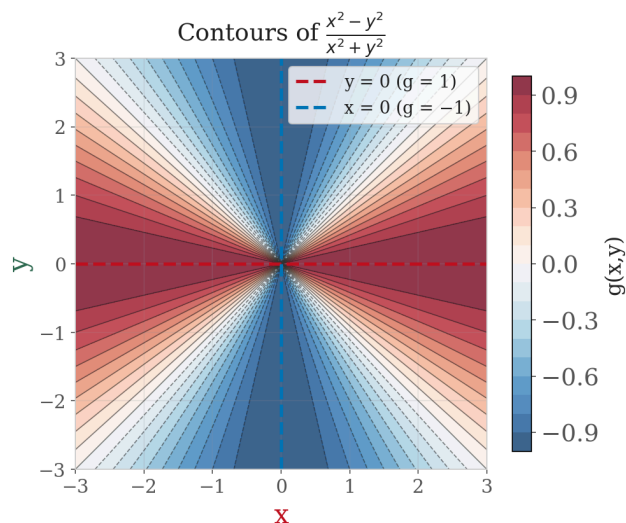
Desmos 3D [https://www.desmos.com/3d]

$$g(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$$

$$g(x, y) = (x^2 - y^2) / (x^2 + y^2)$$



Desmos 3D [https://www.desmos.com/3d]



Desmos 3D [https://www.desmos.com/3d]

f settles down to a single value (1) no matter how you approach the origin.

g gives **different values** depending on your path — the limit **does not exist**.

The Formal Definition of a Multivariable Limit

Definition 1: Limit of a Function of Two Variables

Let f be a function of two variables whose domain D includes points arbitrarily close to (a, b) . We say that the **limit of $f(x, y)$ as (x, y) approaches (a, b) is L** , and write

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

if for every $\varepsilon > 0$ there is a corresponding $\delta > 0$ such that

$$\text{if } (x, y) \in D \text{ and } 0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta \text{ then } |f(x, y) - L| < \varepsilon$$

What does this mean in plain language?

If the limit is going to be L , then you must be able to make $f(x, y)$ as close to L as you want, just by squeezing (x, y) close enough to (a, b) — from **any direction**.

The crucial phrase: **from any direction**. In Calc I, "closeness" meant $|x - a| < \delta$ (an interval). Here, closeness means $\sqrt{(x - a)^2 + (y - b)^2} < \delta$ (a disk).

In practice, we rarely use the ε - δ definition directly. Instead, we develop **strategies** for showing limits exist or don't exist.

Strategy: Showing a Limit Does NOT Exist

If a limit exists, it must be the **same value** no matter which path you take to the point. So to show a limit **does not** exist, we just need to find trouble:

The Two-Path Test

Find two paths to (a, b) that give **different limiting values**. Then the limit does not exist.

Common paths to try:

- Along the x -axis: set $y = 0$
- Along the y -axis: set $x = 0$
- Along $y = x$ (diagonal)
- Along $y = mx$ (any line through the origin)
- Along curves: $y = x^2$, $y = x^3$, etc.

Warning: If two paths give the **same** value, that does **not** prove the limit exists. It just means those two paths agree. You would need to check **all** paths.

Example 1: The Two-Path Test in Action

Example 1

Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Try the paths $y = 0$ and $x = 0$. What value does the function approach along each?

Example 2: When Simple Paths Aren't Enough

Example 2

Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{8x^3y}{x^6 + y^2}$ does not exist.

Try the obvious paths (axes, $y = mx$) first, then try a curved path like $y = x$ -cubed.

Your Turn: Test This Limit

Example 3

Does $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ exist?

Try a path that will obviously give a limit of zero, then see if you can think of path that could do something else.

Continuity of Multivariable Functions

Continuity works just like Calc I — the limit equals the function value at the point.

Definition 1: Continuity at a Point

A function f of two variables is **continuous at** (a, b) if:

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$$

This requires three things (same as Calc I):

1. $f(a, b)$ is defined
2. The limit exists
3. They are equal

Good news: Polynomials, rational functions (where defined), trig functions, exponentials, logarithms, and compositions of continuous functions are all continuous on their domains. Most functions you encounter are continuous wherever they're defined.

Strategy: Showing a Limit DOES Exist

The two-path test can **disprove** a limit, but it can never **prove** one.

To prove a limit exists, we need tools that handle **all paths at once**.

Tool 1: Squeeze Theorem

If $|f(x, y) - L| \leq g(x, y)$ and $g(x, y) \rightarrow 0$ as $(x, y) \rightarrow (a, b)$, then $f(x, y) \rightarrow L$.

Trap the function between bounds that both go to the same value.

Tool 2: Polar Coordinates

Substitute $x = r \cos \theta$, $y = r \sin \theta$. Then $(x, y) \rightarrow (0, 0)$ becomes $r \rightarrow 0^+$.

Apply the Squeeze Theorem to bound the θ -dependent factor and show the expression goes to zero.

Both tools convert the multivariable problem into something we can handle with Calc I techniques.

Example 4: Proving a Limit Exists

Example 4

Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{5xy^2}{x^2 + y^2} = 0$.

Try the Squeeze Theorem using $y^2 \leq x^2 + y^2$. Then try polar coordinates and apply the Squeeze Theorem to the result.

Your Turn: Prove or Disprove

Example 5

Does $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$ exist? If so, find its value.

Use either the squeeze theorem or polar coordinates.

Homework

Section 14.2: 5, 7, 9, 15, 17, 21, 22, 23, 25, 29, 41, 43, 47, 49, 51